

Enhancing Health Management Through Wearable Devices

Team RCJ

Members: Hsueh-Ta Lu, Tung-Ying Lee, Po-Han Chen, Hsin-Ying Tsai

WATCHER







Motivation & Problem Statement

Current Challenge

Wearable devices like Apple Watch and Fitbit collect extensive health data, yet users struggle to interpret and apply the information effectively.

Our Focus

Bridge the gap between raw data and meaningful insights through intuitive design and AI-driven analysis to enhance user engagement and health outcomes.

Access vs. Action

Users have rich health data but often miss critical insights needed to take proactive wellness actions.

Proposed Solution:

AI-powered interface that simplifies data and offers personalized, actionable insights.

User Research Insights: Enhancing Wearable Health Devices

Research Methods

- **Qualitative Interviews**

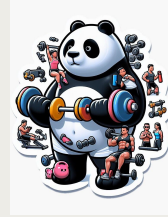
- 5 participants (age 22–35)
- Fitness enthusiasts & casual users
- Conducted in-person & via Zoom

- **Quantitative Surveys**

- Google Forms
- 10–20 respondents
- Usage patterns, frustrations, and feature preferences



Primary Users



- Fitness enthusiasts improving daily health

- Chronic condition patients needing continuous monitoring



Secondary Users



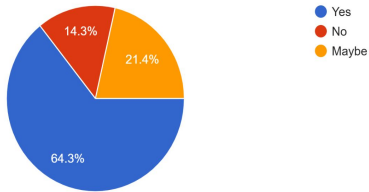
- Researchers studying digital health patterns

- Healthcare providers offering remote care



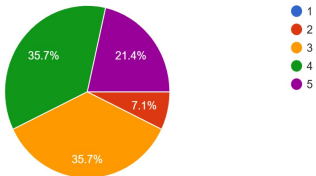
Do you find the data provided by the wearable device easy to understand?

14 則回應



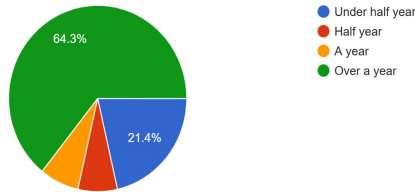
On a scale of 1-5, how satisfied are you with the health tracking features of your wearable device? (Linear Scale: 1 = Very Unsatisfied, 5 = Very Satisfied)

14 則回應



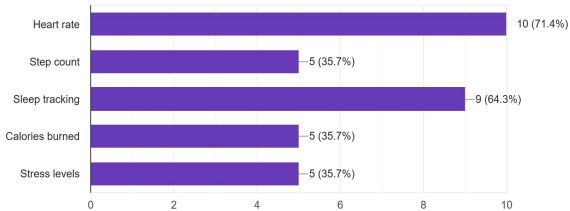
How long have you been using wearable devices for health tracking?

14 則回應



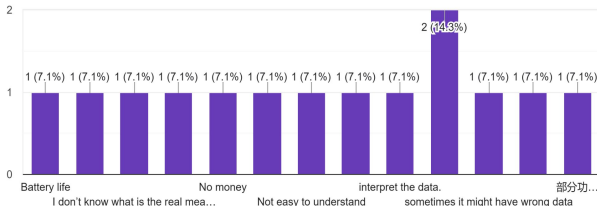
Which health metrics do you find most useful on your wearable device?

14 則回應



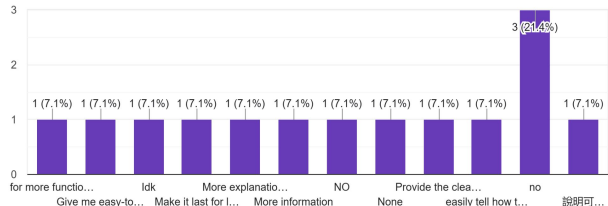
What challenges do you face when using health tracking features?

14 則回應



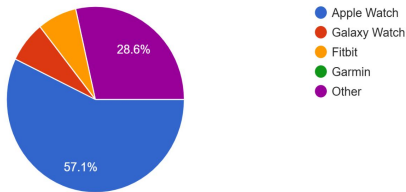
What improvements would you suggest for wearable health management devices?

14 則回應



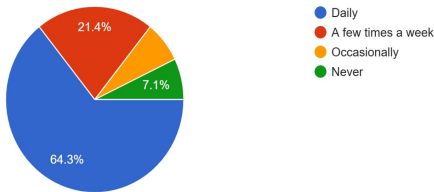
Which wearable devices have you used?

14 則回應



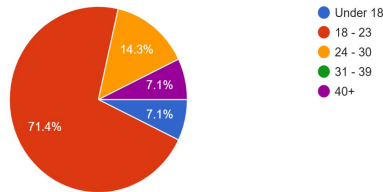
How often do you use the health and fitness tracking features on your wearable device?

14 則回應



What is your age group?

14 則回應





Core Insights

- **Data Overload:** Complex metrics like HRV confuse users.
- **Need for Actionable Tips:** Users want clear, personalized guidance.
- **Low Engagement:** Casual users lose interest over time.
- **Simplicity Matters:** Clean, goal-based interfaces are preferred.



Quantitative Highlights

- **Understanding:** 3/5 users struggled with HRV/stress data.
- **Engagement:** Only 2/5 used data regularly; others wanted nudges.
- **Notifications:** Mixed feedback—some liked them, others felt fatigued.

71.4%

Age Group

Majority of users aged 18–23

64.3%

Daily Usage

Tracked heart rate and sleep most frequently

14.3%

Data Comprehension Challenges

Users struggling to understand health data

From HMW Questions to Design Ideas

Actionable Suggestions from Data

How might we turn raw health metrics into clear, practical recommendations users can trust and follow?

Simplify Wearable Data

How might we make wearable health data easier to understand and use to motivate users effectively?

Personalized Insights

How might we provide tailored health insights based on observed trends, such as the impact of caffeine on heart rate?



AI-Generated Recommendations

Automated health action suggestions tailored to individual data patterns enhance user decision-making.



Summary Cards

Daily and weekly cards with clear visuals help users grasp overall health trends at a glance.



Personalized Lifestyle Tips

Insights based on heart rate and sleep trends guide tailored behavior changes to improve wellness.

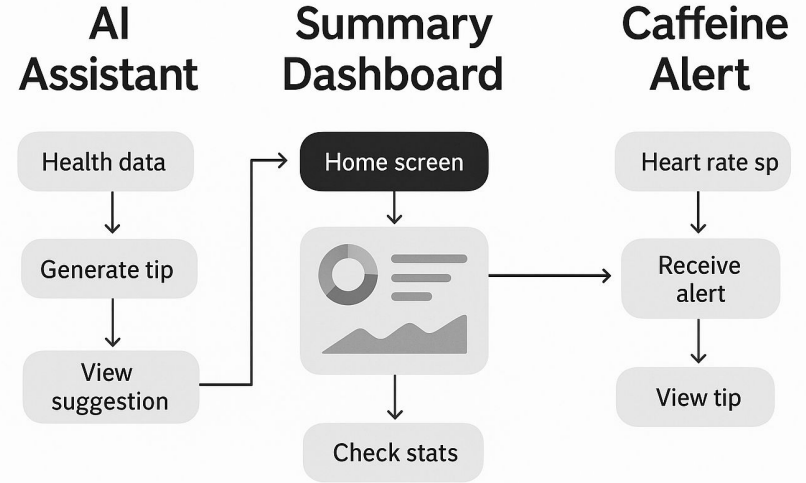
Final Selected Concepts

AI Assistant

Converts complex metrics into simple, personalized health tips, building trust and understanding.

Simplified Summary Dashboard

Weekly health overviews with intuitive visuals and explanations support ongoing user engagement.



Caffeine-Aware Alerts

Custom notifications provide timely tips when heart rate spikes potentially related to caffeine intake.

Low-Fidelity Phase Outcomes

User Interest Validated

Users preferred simple, actionable insights instead of complex charts.

AI-Supported Feedback

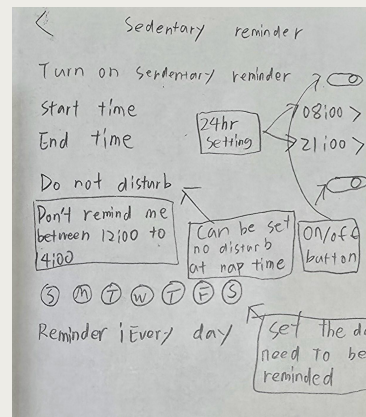
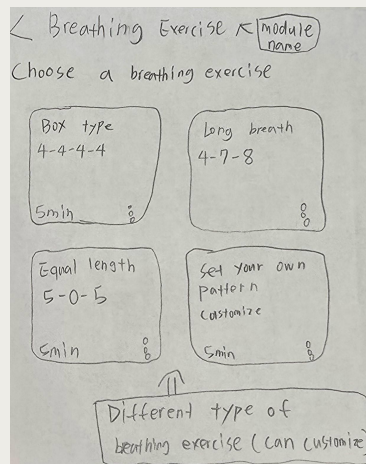
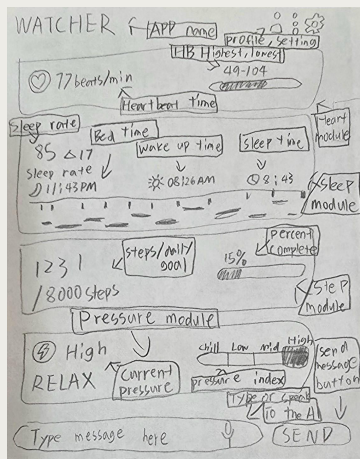
Feedback loops powered by AI enhanced user engagement and understanding.

Unified Health Dashboard

The interface consolidated health metrics seamlessly, fostering user confidence.

This validation encourages us to move towards interactive, medium-fidelity design phases with enhanced flows and visual hierarchy.



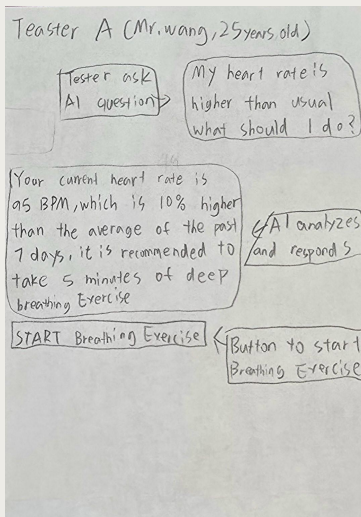


Revised prototype

Improvement points

More detail on AI analyzes and responds

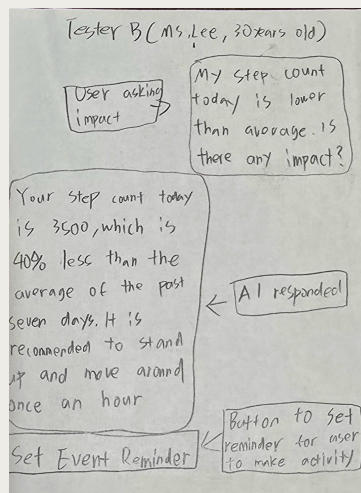
Make AI responses more intuitive and button design more in line with user needs.



Feedback:

Like the "Start Breathing exercise" button, feels very intuitive.

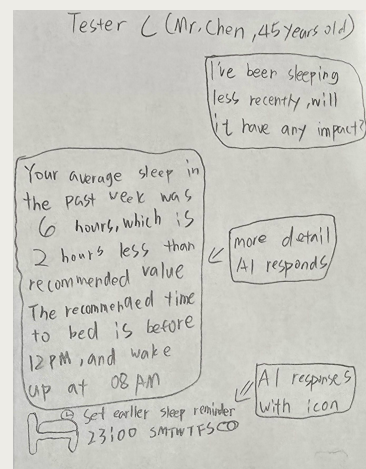
Hope to add further analysis on "why does the heart rate become high"?



Feedback:

Like the data comparison and feel that this suggestion is well-founded.

Hope AI can provide "alternative solutions", such as simple indoor sports.



From Lo-Fi to Mid-Fi: Key Updates and Testing Insights



Simple Onboarding

Improved early user engagement through streamlined sign-up and data input.



Mood + AI Synergy

A combination of mood tracking and AI feedback was perceived as both informative and enjoyable.



Visual Clarity

Clear visuals boosted user trust and increased retention potential.

Color & Branding

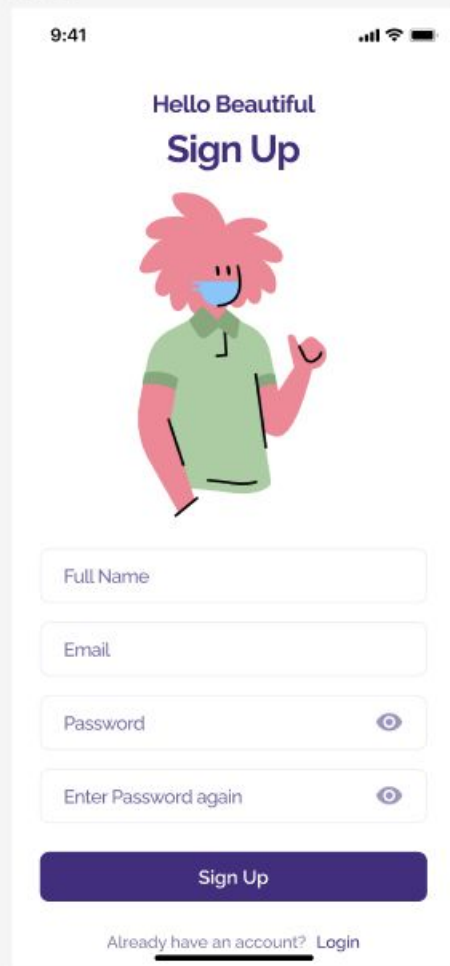
Introduced a consistent purple-based palette with soft visuals to reinforce calmness and trust.

Typography & Layout

Enhanced text hierarchies and spacing for greater readability and organization.

UI Components

Unified button sizes, icon styles, and layout grids to create a cohesive interface.



HR1

9:41

 **Your result**

Your Heart Data

 Pulse

 Heart Score

 HRV

XX
BPM

60
/100

90
MS

Comment about heart comment about heart comment about heart

Stress Mild 10%

Energy ????? 75%

Comment about heart comment about heart comment about heart

What is your mood?

SAVE

HR2

9:41

 **Your result**

Comment about heart comment about heart comment about heart

What is your mood?

What have you been up to?

 Relaxing

 Sleep

 After sleep

Comment

Q W E R T Y U I O P

A S D F G H J K L

↑ Z X C V B N M ↵

123 space return

HR3

9:41

 **Your result**

Comment about heart comment about heart comment about heart

Stress Mild 10%

Energy ????? 75%

Comment about heart comment about heart comment about heart

What is your mood?

What have you been up to?

 Relaxing

 Sleep

 After sleep

Comment

SAVE

Outcome of High-Fidelity Prototyping

Validated Usability & Design

- Users navigated with minimal confusion
- Realistic interactions confirmed intuitive UI

Identified Subtle Issues Early

- Surfaced minor usability problems
- Enabled early fixes before development

Informed Final Design Decisions

- Built confidence in product readiness
- Reduced risk of costly post-release changes

Aligned with User Expectations

- Iterations refined based on user needs
- Ensured UI choices met actual use cases

9:41



My Health Overview



Apr 14 - Apr 20

Heart Rate
68



HRV
65%



Stress
Low



Sleep
7h 20m



Energy Score 82

Achievements
Logged all week!

Your stress decreased compared to last week – keep it up! 🍌

SHARE



9:41



Your weekly report is ready!

You've done great this week:
Average Sleep: 7h 20 m
HRV improved ✓
Stress decreased ✓



Consistency is key!
Keep logging daily.

COPY LINK

SHARE TO ...



9:41



Settings



Account



Connected Devices



Notifications



My Goals



Appearance



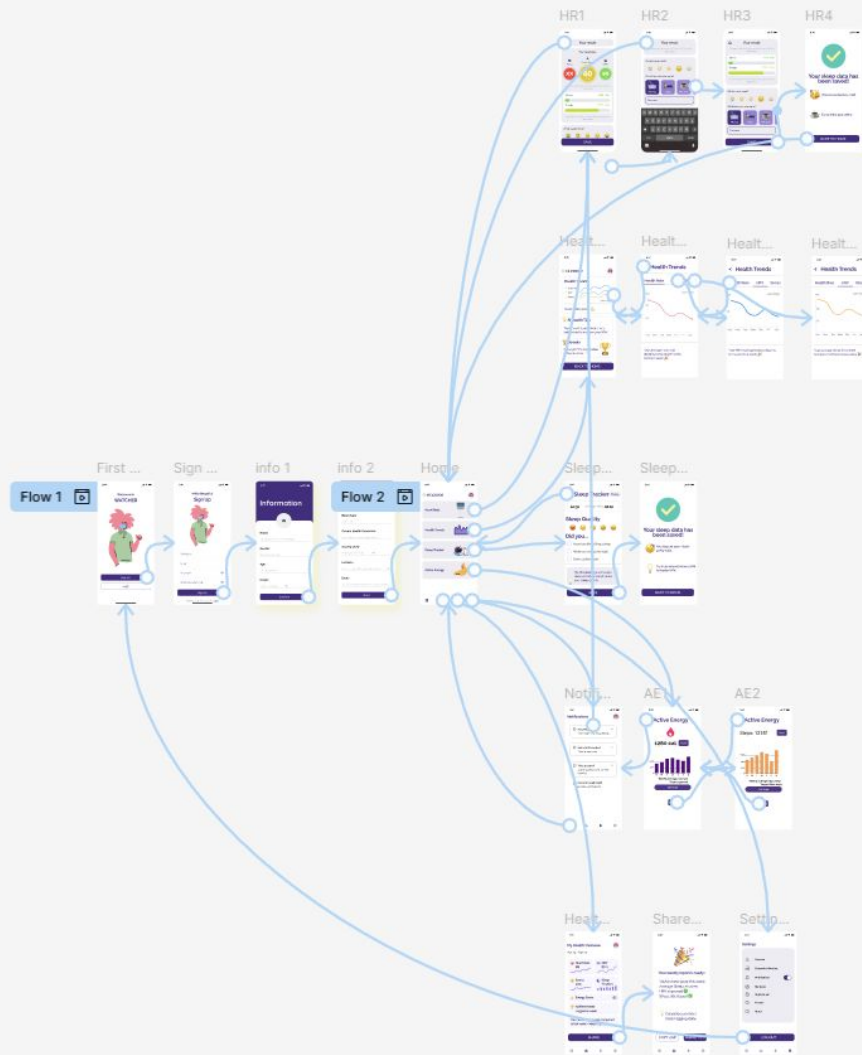
Privacy



About

LOG OUT





Final Reflections & Key Learnings from User-Centered Design

1

User-Centered Design Is Essential

Friendly and non-judgmental AI responses helped users feel more comfortable and engaged.

2

Designed for Clarity and Action

Transformed raw metrics into simple, digestible insights that users could easily understand and apply.

3

Built Through Feedback

Iterative testing and user feedback refined both visuals and functionality.

4

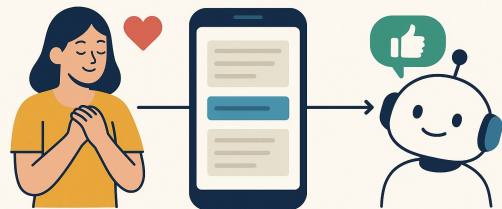
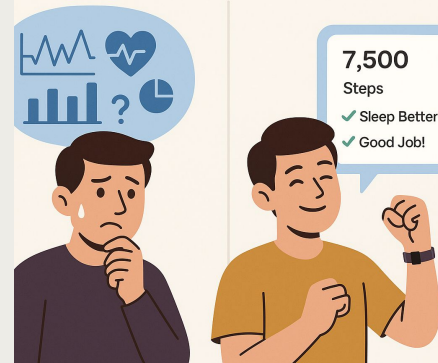
Celebrate Small Wins

Small, daily wins kept users motivated and encouraged sustainable health habits.

5

Details Matter

Micro-interactions like hover effects, emojis, and tone made the experience feel human—not robotic.



Empathy

Understanding user needs

Clarity

Making information clear

Supportive AI

Providing helpful guidance